

1. Solution Description: Chain IIoT

Chain IIoT is a lightweight, all-in-one **cloud-edge automation platform** positioned as a practical and significantly lower-cost alternative to classical industrial automation stacks (PLC + industrial HMI + protocol gateway).

Core capabilities

- **All-in-one edge controller**

Integrates control logic, hybrid field networking, local operator interface and gateway functions into a single device. Because the platform is **web-native**, the edge controller itself can be realized on **commodity hardware** rather than expensive industrial-grade controllers.

- **Hybrid field connectivity**

Native simultaneous support for **Modbus RTU (RS-485 multi-drop)** and **transparent LoRa**, allowing existing wired devices and new wireless sensors/actuators to be orchestrated together without extra gateways.

- **Local autonomy first**

Real-time control loops, sequencing, interlocks and safety logic execute on the edge and continue during cloud or network outages.

- **Fully web-native architecture**

The entire stack — edge runtime, engineering/configuration, visualization and cloud interaction — is web-native (browser + app-script + on-device ML). This has a decisive cost and accessibility impact:

- The edge controller itself can run on commodity hardware.
- Local and remote operator interfaces use ordinary browsers.
- Large dashboards can use ordinary **Android / smart TVs** (or any browser-capable screen) in kiosk mode.
- Most related devices in the architecture can be **commodity devices** instead of expensive industrial HMIs, industrial PCs or proprietary panels.
- No proprietary PLC programming environments or runtime licenses are required.

- **Cloud-based business processing & business-driven edge automation**

The cloud performs higher-level **business processing** (multi-site optimization, energy/tariff logic, service-level policies, scheduling, analytics, outcome tracking) and actively **drives the edge** by pushing business rules, setpoints and control strategies. The edge remains responsible for real-time execution while becoming an instrument of business objectives. This enables true Automation-as-a-Service and outcome-based models.

- **Cloud-managed facility definitions**

Complete predefined facility packages (network topologies, scripting rules, ML models, setpoints, visualization layouts) are stored and versioned in the cloud. During edge deployment or later in production, operators can simply **pick and switch** the appropriate definition — turning repeated similar sites into configuration actions rather than re-engineering projects.

- **Clean control / visualization split**

The edge device owns all control. Large dashboards are pure display surfaces and can use ordinary commodity screens.

Positioning

Chain IIoT targets the broad mid-market where classical industrial stacks are economically or operationally excessive. By making the **entire architecture web-native**, it allows most components — including the edge controller itself — to be commodity devices, dramatically lowering cost while still delivering reliable local control and modern cloud-driven, business-oriented automation.

2. Comparison vs Classical Industrial Stacks

(Siemens, Schneider Electric, Rockwell Automation and equivalents)

Dimension	Classical Industrial Stack	Chain IIoT
Core hardware	Industrial PLC(s) + industrial HMI + gateway(s)	All-in-one edge controller that can run on commodity hardware
Field networks	PROFINET / EtherNet/IP / Modbus / BACnet (extra gateways often needed)	Native Modbus RTU + transparent LoRa
Control logic	Vendor PLC languages + specialized IDEs	Fully web-native (app-script + edge ML)
Facility / project reuse	Templates possible; most sites still need significant engineering	Cloud library of complete definitions — pick & switch at deploy or runtime

Local autonomy	Strong	Strong
Cloud role	Mainly telemetry, remote access, optional SCADA/analytics	Cloud-based business processing that actively drives edge automation with business policies
Operator / large display	Expensive industrial HMI panels + dedicated industrial monitors	Ordinary browsers + commodity Android / smart TVs (or any screen) in kiosk mode
Device cost philosophy	Industrial-grade devices throughout	Most devices can be commodity (including the edge controller itself)
Cost profile	High CapEx + specialized engineering + licenses	Significantly lower hardware & integration cost
Deployment speed	Days to months	Faster, especially with pre-defined packages
Skill barrier	Specialized OT / PLC engineers	Lower — web-configurable + selectable packages
Service model fit	Primarily product / project sales	Designed for Automation-as-a-Service and business-outcome models
Best scale fit	Medium to very large / high-criticality processes	Small to mid-scale, multi-site, cost-sensitive, retrofit

Key differentiators of Chain IIoT

1. Fully web-native architecture allows the edge controller and most related devices to be commodity hardware instead of expensive industrial devices.
2. Cloud-managed facility definitions enable rapid, low-engineering replication of proven configurations.
3. The cloud performs real **business processing** and uses the results to drive edge automation, rather than treating the edge as a simple data source or remote I/O.

3. Domains They Apply Respectively

Classical stacks remain preferable when:

- Process criticality or regulations demand certified, highly deterministic PLC platforms with extensive functional safety (SIL) options
- Installations are large, complex and highly customized
- There is already deep investment in a specific vendor ecosystem
- Hard real-time motion control or complex continuous process control is required

- Customer specifications explicitly mandate industrial PLC platforms

Typical domains: Heavy process industries, large discrete manufacturing, major critical infrastructure, large hospitals, large campus BMS.

Chain IIoT is preferable when:

- Classical stacks are economically or operationally excessive for the scale of the site
- Rapid replication of similar facilities is needed
- Hybrid Modbus + LoRa connectivity and easy wireless expansion are valuable
- Multi-site operators want central **business processing** and policy management with local autonomy
- Automation-as-a-Service or outcome-based commercial models are desired
- Visualization and even the controller itself can use commodity devices
- Engineering resources are limited and configurable packages are preferred

The two approaches are **complementary**.

4. Addressable Markets for Chain IIoT

(with typical problems and why Chain IIoT fits)

Primary Markets

1. Controlled Environment Agriculture (CEA)

Greenhouses, high/low tunnels, vertical farms

- **Typical problems:** Full industrial climate computers and PLC systems are too expensive for many mid-sized growers; partial or manual control is common; multi-site recipe and climate consistency is hard; wireless sensor expansion is costly with classical gateways.
- **Why Chain IIoT fits:** Affordable commodity-based edge control + hybrid Modbus/LoRa; cloud facility packages (e.g. “tomato high-tunnel”, “leafy-greens rack”) can be picked and applied quickly; cloud business processing enables multi-site optimization, energy/tariff logic and Automation-as-a-Service offerings; ordinary screens suffice for dashboards.

2. Smart Care / Elderly Care / Senior Living

Assisted living, nursing homes, memory care, long-term care

- **Typical problems:** Budgets and OT expertise are limited; classical BMS is overkill; environmental comfort, simple sensing and status visibility are needed; many facilities already have Android TVs for digital signage.

- **Why Chain IIoT fits:** Low-cost commodity edge control of HVAC and sensors; LoRa for easy wireless sensing; existing Android TVs become natural dashboards; cloud business processing supports multi-site care operators and managed comfort/safety services.

3. Smart Buildings (small-to-medium commercial)

Offices, retail, schools, clinics, smaller hotels

- **Typical problems:** Full BMS is frequently over-specified and too expensive; multi-building portfolios lack affordable central policy and energy optimization; retrofit of mixed Modbus devices is cumbersome.
- **Why Chain IIoT fits:** Lightweight multi-protocol edge control on commodity hardware; cloud business processing for portfolio energy/tariff optimization; facility definitions accelerate repeated building types; ordinary screens for facility dashboards.

4. Smart Communities

Gated communities, apartment complexes, senior-oriented housing clusters

- **Typical problems:** Shared systems need distributed control; classical industrial stacks are rarely justified; community operators want central visibility and service packages.
- **Why Chain IIoT fits:** Distributed commodity edge nodes with local autonomy + cloud orchestration; facility packages for standard community zones; existing TVs in clubhouses/lobbies as display surfaces; business-driven automation for shared-resource optimization and resident service models.

5. Aquaculture

Fish/shrimp farms, recirculating systems

- **Typical problems:** Water quality, pumps, aeration and feeders need reliable control; industrial PLC solutions are often too costly for mid-sized farms; multi-site consistency and remote service are desired.
- **Why Chain IIoT fits:** Very similar operational profile to CEA; hybrid connectivity; cloud facility packages and business processing for multi-farm management and service offerings; commodity hardware keeps cost low.

6. Cold Chain & Food Storage

Cold rooms, refrigerated warehouses, produce storage

- **Typical problems:** Continuous temperature/humidity monitoring and simple control are critical; multi-site compliance and audit trails are required; classical SCADA is expensive for mid-sized or distributed facilities.
- **Why Chain IIoT fits:** Reliable local control on commodity edge hardware + LoRa retrofit sensors; cloud business processing for multi-site compliance, alerts and reporting; rapid deployment via facility definitions.

Secondary / Adjacent Markets

- **Open-field & broader agriculture** (irrigation, livestock, grain storage)
Problems: Long-range sensing, pump/valve control, multi-farm coordination.
Fit: LoRa + Modbus + cloud multi-farm business logic on low-cost edge hardware.
 - **Water & wastewater (small/mid-scale)**
Problems: Classical SCADA too expensive for smaller systems; remote managed services attractive.
Fit: Hybrid networks + local autonomy + cloud business processing for smaller utilities.
 - **Energy edge** (solar + storage, microgrids, EV charging clusters)
Problems: Modbus-heavy equipment, tariff optimization, aggregation.
Fit: Native Modbus + cloud business processing for optimization and service models.
 - **Hospitality & multi-site retail**
Problems: Multi-site HVAC/energy consistency, cost pressure, existing TVs.
Fit: Central business policies + edge autonomy + commodity dashboards.
 - **Light industrial / small manufacturing**
Problems: Full industrial stacks excessive for many SMEs.
Fit: Affordable commodity edge control + web-native configuration.
 - **Education & campus facilities**
Problems: Multi-building, budget-constrained environments.
Fit: Lightweight distributed control + central cloud management.
 - **Municipal / smart-city edge**
Problems: Fragmented use cases, preference for incremental deployment.
Fit: LoRa + commodity edge autonomy for phased, lower-cost roll-outs.
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Summary

Chain IIoT delivers reliable edge control, hybrid field networking, cloud-managed facility definitions, and **cloud-based business processing that drives business-oriented edge automation**. Because the **entire architecture is web-native**, most related devices — including the edge controller itself — can be commodity hardware rather than expensive industrial devices. This combination makes Chain IIoT the practical choice for the broad mid-market of CEA, smart care, smart buildings, smart communities and related cost-sensitive, multi-site domains where classical automation has been economically or operationally out of reach.